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ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN-Shift-1** **Date: 21-03-2026**

Time: 09.00AM to 12.00PM **GTM-25** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

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21-03-2026_Sr.S60(N&S), Elite, IAS & LIIT_Jee-Main-Shift-1_GTM-25_Test Syllabus

MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. $\tan^{-1}\left(\frac{4}{7}\right)+\tan^{-1}\left(\frac{4}{19}\right)+\tan^{-1}\left(\frac{4}{39}\right)+\tan^{-1}\left(\frac{4}{67}\right)+\ldots\ldots\ldots\infty=$ _____

1) $\cot^{-1}2$ **2)** $\frac{\pi}{4}$ **3)** $\tan^{-1}2$ **4)** $\frac{\pi}{2}$

2. Equation of the common tangent touching the circle $(x-3)^2+y^2=9$ and the parabola $y^2=4x$ above the x - axis is

1) $\sqrt{3}y=3x+1$ **2)** $\sqrt{3}y=-(x+3)$ **3)** $\sqrt{3}y=(x+3)$ **4)** $\sqrt{3}y=-(3x+1)$

3. The values of a for which the expression $\frac{ax^2+3x-4}{a+3x-4x^2}$ can assume all real values for real x, lie in the interval

1) $a\le 1$ *or* $a\ge 7$ **2)** $a\ge 1$ *or* $a\le 7$ **3)** $1\le a\le 7$ **4)** $a<1$ *or* $a>7$

4. If the roots of $x^4-ax^3+bx^2-cx+d=0$ are real and positive, then the minimum value of $\frac{(a+b+c)}{2\sqrt{d}}$, is

1) 5 **2)** 6 **3)** 7 **4)** 8

5. The reflection of the point $A(1,0,0)$ in the line $\frac{x-1}{2}=\frac{y+1}{-3}=\frac{z+10}{8}$ is

1) $(3,-4,-2)$ **2)** $(5,-8,-4)$ **3)** $(1,-1,-10)$ **4)** $(2,-3,8)$

6. Let AC be an arc of a circle, subtending a right angle at the centre O. The point B divides the arc AC in the ratio $1:2$. If $\overline{OA}=\bar{a}$ and $\overline{OB}=\bar{b}$, then $\overline{OC}$ in terms of $\bar{a}$ and $\bar{b}$ is

1) $\bar{b}-\sqrt{3}\,\bar{a}$ **2)** $2\bar{b}-\bar{a}$ **3)** $2\bar{b}-\sqrt{3}\,\bar{a}$ **4)** $\bar{b}-\bar{a}$

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7. If $[\ ]$ denotes the greatest integer function then $\int_0^{[x]} \frac{2^x}{2^{[x]}}dx =$

1) $\frac{[x]}{\log 2}$ **2)** $\frac{-[x]}{\log 2}$ **3)** $\frac{-[2x]}{\log 2}$ **4)** $\frac{[2x]}{\log 2}$

8. The number of real values of t such that the system of homogeneous equations $tx+(t+1)y+(t-1)z=0;\ (t+1)x+ty+(t+2)z=0;\ (t-1)x+(t+2)y+tz=0$ Has non-trivial solutions, is

1) 3 **2)** 2 **3)** 1 **4)** 4

9. If $f(x)=\begin{cases} x[x] & , \quad 0\le x<2 \\ (x-1)[x], & 2\le x\le 3 \end{cases}$, where $[.]$ denotes the greatest integer function, then

1) Both $f'(1)$ and $f'(2)$ do not exist **2)** $f'(1)$ exists but $f'(2)$ does not exist

3) $f'(2)$ exists but $f'(1)$ does not exist **4)** Both $f'(1)$ and $f'(2)$ exists

10. The area common to the circle $x^2+y^2=16a^2$ and the parabola $y^2=6ax$ is

1) $4a^2(8\pi-\sqrt{3})$ **2)** $\frac{4a^2(4\pi+\sqrt{3})}{3}$ **3)** $\frac{4a^2(8\pi-\sqrt{3})}{5}$ **4)** $\frac{4a^2(\pi+\sqrt{3})}{3}$

11. $1-\frac{1}{5}+\frac{1.4}{5.10}-\frac{1.4.7}{5.10.15}+.....\infty =$

1) $\sqrt[3]{5}$ **2)** $\frac{1}{2}\sqrt[3]{5}$ **3)** $\left(\frac{5}{2}\right)^{\frac{1}{3}}$ **4)** $\frac{1}{2}\left(\frac{5}{2}\right)^{\frac{1}{3}}$

12. Let $f:R\to R$ be defined by $f(x)=\begin{cases} \alpha+\frac{\sin[x]}{x} & ;\ if\ x>0 \\ 2 & ;\ if\ x=0 \\ \beta+\left[\frac{\sin x-x}{x^3}\right] & ;\ if\ x<0 \end{cases}$

where $[y]$ denotes the integral part of y. If f is continuous at $x=0$, then $\beta-\alpha=$

1) -1 **2)** 1 **3)** 0 **4)** 2

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13. If f is a polynomial function satisfying $2+f(x)f(y)=f(x)+f(y)+f(xy)\ \ \forall x,y\in R$ and If $f(2)=5,\ f(1)\neq 1$ then the value of $f\left(f(2)\right)=$

1) 25 **2)** 16 **3)** 26 **4)** 14

14. Match the differential equations in List I to their integrating factors in List II

List I Differential Equation	**List II** Integrating Factor
i) $(x^3+1)\frac{dy}{dx}+x^2y=3x^2$	a) x^3
ii) $x^2\frac{dy}{dx}+3xy=x^6$	b) $(x^3+1)^2$
iii) $(x^3+1)^2\frac{dy}{dx}+6x^2(x^3+1)y=x^2$	c) $(x^2+1)^2$
iv) $(x^2+1)\frac{dy}{dx}+4xy=\ln x$	d) x^2+1
	e) $(x^3+1)^{1/3}$
	f) $(x^3+1)^{1/2}$

The correct match is

1) i-d ;ii-a ; iii-b ; iv-c **2)** i-e ;ii-a ; iii-b ; iv-c

3) i-e ;ii-b ; iii-c ; iv-f **4)** i-e ;ii-a ; iii-c ; iv-d

15. **Statement – I:** Let N denote the set of all natural numbers and R be the relation on $N\times N$ defined by $(a,b)\ R(c,d)$ if $ad(b+c)=bc(a+d)$, then R is an equivalence relation.

Statement – II: Let A be a set. A relation R in A is said to be an equivalence relation if R is reflexive, symmetric and transitive in A.

1) Statement I is true, Statement II is true, Statement II is correct explanation for Statement I.

2) Statement I is true, Statement II is true, Statement II is not correct explanation for Statement I.

3) Statement I is true, Statement II is false.

4) Statement I is false, Statement II is true.

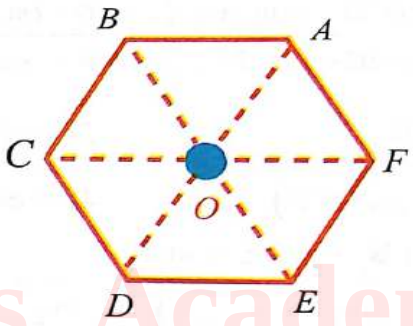
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16. $\int \frac{ax^2-b}{x\sqrt{c^2x^2-\left(ax^2+b\right)^2}}dx =$

1) $\sin^{-1}\left(\frac{ax+b/x}{c}\right)+k$ **2)** $\sin^{-1}\left(\frac{ax^2+b/x^2}{c}\right)+k$

3) $c\cos^{-1}\left(\frac{ax^2+b/x^2}{c}\right)+k$ **4)** $\cos^{-1}\left(\frac{ax^2-b/x^2}{c}\right)+k$

17. Complex numbers can be represented as a position vector, if centre of a regular hexagon is at $(0,0)$ and one vertex A is $\sqrt{3}+i$



For the above figure find $\overline{OB}$.

1) $2e^{\frac{i\pi}{2}}$ **2)** $2e^{\frac{i\pi}{6}}$ **3)** $2e^{\frac{i\pi}{3}}$ **4)** $2e^{\frac{i\pi}{10}}$

18. A channel $27\ m$ wide falls at a right angle into another channel $64\ m$ wide. The least length of the log that can be floated along this system of channels is

1) 120 **2)** 125 **3)** 100 **4)** 110

19. Three '1×1' squares of a chess board having 8×8 squares are chosen at random, the chance that 2 are of one colour and 1 is of different colour is

1) $\frac{^{32}C_3}{^{64}C_3}$ **2)** $\frac{^{16}C_3}{^{64}C_3}$ **3)** $\frac{2\times{}^{32}C_2\times{}^{32}C_1}{^{64}C_3}$ **4)** 1

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20. Sum of the series $\sum_{r=0}^{n}(-1)^r\,{}^nC_r\left[i^{5r}+i^{6r}+i^{7r}+i^{8r}\right]$ is, where i is the non-real 4th root of unity.

1) 2^n **2)** $2^{\frac{n}{2}+1}$ **3)** $n^n+2^{\frac{n}{2}+1}$ **4)** $2^n+2^{\frac{n}{2}+1}\cos\frac{n\pi}{4}$

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. Number of points having position vector $ai+bj+ck$, where $a,b,c\in\{1,2,3,4,5\}$ such That $2^a+3^b+5^c$ is divisible by 4 is

22. Mean deviation of 390, 400, 400, 410, 410, 420, 420, 430, 430, 440, 440, 450 through median is

23. Number of solutions of $2\tan^{-1}(\tan x)=6-x$ is

24. $A(-3,2),B(2,5)$ be two points. If the point $P(0,K)$ on y-axis, such that $|PA-PB|$ is greatest, then $\frac{K-1}{5}$ is equal to

25. ABCD is a quadrilateral with $\overrightarrow{AB}=\bar{a},\overrightarrow{AD}=\bar{b},\ and\ \overrightarrow{AC}=2\bar{a}+3\bar{b}$. If its area is α times the area of the parallelogram with AB,AD as adjacent sides, then 2α is equal to

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PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. **Assertion (A)**: A non-polar material do not have any permanent electric dipole moment, but when placed in external electric field it develops electric dipole moment.

Reason (R): when a non-polar material is placed in an electric field, the centre of the positive charge distribution of its individual atom or molecule does not coincide with the centre of the negative charge distribution.

In the light of above statements, choose the most appropriate answer from the options given below

1) Both A and R are correct and R is the correct explanation of A

2) Both A and R are correct and R is not the correct explanation of A

3) A is correct but R is not correct.

4) A is not correct but R is correct

27. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): In a static magnetic field, the velocity of a charged particle changes only in direction.

Reason (R): Moving charged particle experiences magnetic force in magnetic field which is perpendicular to its direction of velocity.

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true and R is not the correct explanation of A

3) A is true but R is false.

4) A is false but R is true.

28. Given below are two statements:

Statement I: An AC circuit undergoes electrical resonance if it contains both capacitor as well as inductor.

Statement II: If a pure inductor is connected across alternating voltage source then instantaneous voltage (v) instantaneous current (i) and inductive reactance (x_L) are related as $i = \frac{v}{(X_L)}$.

In the light of above statements choose the correct answer from the options given below.

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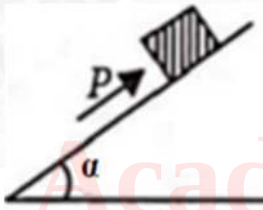


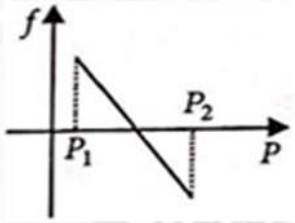
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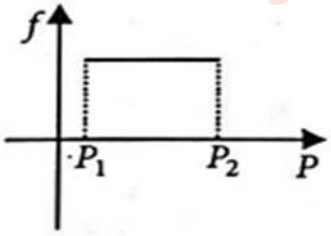
1) Statement I is correct but statement II is incorrect.
2) Statement I is incorrect but Statement II is correct.
3) Both Statement I and Statement II are incorrect.
4) Both Statement I and Statement II are correct.

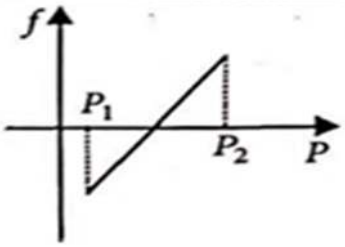
29. Statement-I: Due to rapid pumping of air inside the tyre is hotter than atmospheric air.
Statement-II: Adiabatic process occurs at very high rate
1) Statement-I is true; Statement-II is false.
2) Statement-I is false; Statement-II is true.
3) Statement-I & Statement-II are true.
4) Statement-I & Statement-II are false.

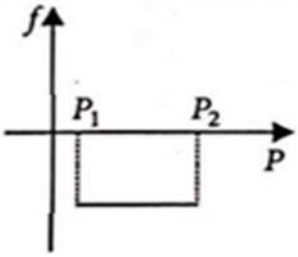
30. Block of mass M is on inclined plane of an angle α. The coefficient of friction between the block and plane is μ and $\tan\alpha > \mu$. Block is held stationary by applying a force P parallel to the plane. Direction of force pointing up the plane is taken to be positive. As P is varied from $P_1 = mg(\sin\alpha - \mu\cos\alpha)$ to $P_2 = mg(\sin\alpha + \mu\cos\alpha)$, the frictional force f versus P graph looks like:



1)


2)


3)


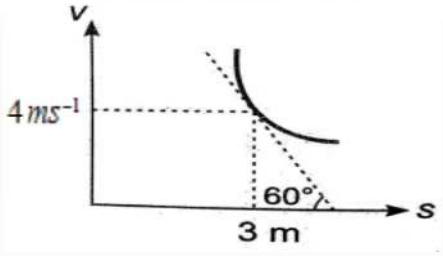
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31. A particle is moving along a straight line whose velocity-displacement graph is as shown in the figure. What is the magnitude of acceleration when displacement is $3m$?

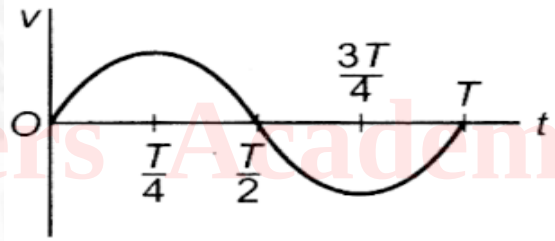


1) $4\sqrt{3}\ ms^{-2}$ **2)** $3\sqrt{3}\ ms^{-2}$ **3)** $\sqrt{3}\ ms^{-2}$ **4)** $\frac{4}{\sqrt{3}}\ ms^{-2}$

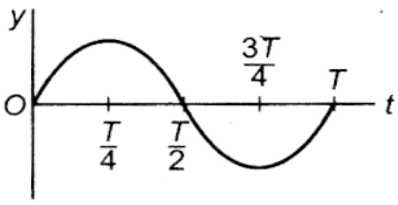
32. If n main scale divisions coincide $(n+2)$ Vernier scale divisions. The least count of Vernier calipers (in cm), when each centimetre on the main scale is divided into five equal parts will be (in cm).

1) $\frac{2n}{n+2}$ **2)** $\frac{5}{n+1}$ **3)** $\frac{2}{5(n+2)}$ **4)** $\frac{5}{n+2}$

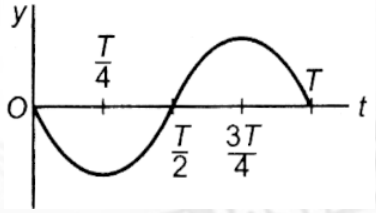
33. Velocity of a particle executing SHM varies with time as shown in Fig. Variation of displacement of the particle from the mean position with time will be as in:



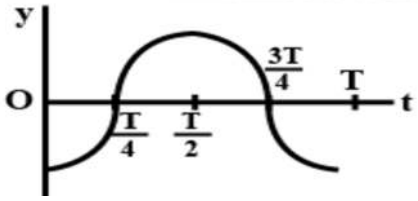
1)



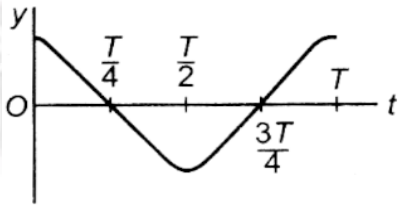
2)



3)



4)



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34. A particle free to move along the x -axis has potential energy given by $U(x)=k\left[1-\exp\left(-x^2\right)\right]$ for $-\infty \le x \le +\infty$, where k is a positive constant of appropriate dimensions. Then

1) At points away from the origin, the particle is in unstable equilibrium.

2) For any finite non-zero value of x , there is a force directed away from the origin.

3) If its total mechanical energy is $\frac{k}{2}$, it has its minimum kinetic energy harmonic.

4) For small displacements from $x=0$, the motion is simple harmonic.

35. A sphere and a cube both made of copper have equal volumes and are blackened. These are heated to same temperature and are allowed to cool under same surroundings. The ratio of their rates of loss of heat is

1) 1 : 1 **2)** $\left(\frac{\pi}{6}\right)^{1/3}$ **3)** $\left(\frac{6}{\pi}\right)^{1/3}$ **4)** $\left(\frac{\pi}{6}\right)^{1/2}$

36. A cavity of radius $\frac{R}{2}$ is made inside a solid sphere of radius R . The centre of the cavity is located at a distance $\frac{R}{2}$ from the centre of the sphere. The gravitational force on a particle of mass m at a distance $\frac{R}{2}$ from the centre of the sphere on the line joining both the centres of the sphere and the cavity is (opposite to the centre of the cavity)

[Here $g=\frac{(GM)}{R^2}$, where M is the mass of the sphere]

1) $\frac{mg}{2}$ **2)** $\frac{3mg}{8}$ **3)** $\frac{mg}{16}$ **4)** $\frac{3mg}{5}$

37. A solid metallic sphere of radius r is allowed to fall freely through air. If the frictional resistance due to air is proportional to the cross – sectional area and to the square of the velocity then the terminal velocity of the sphere is proportional to which of the following?

1) r^2 **2)** r **3)** $r^{3/2}$ **4)** $r^{1/2}$

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38. Poisson ratio of certain material is 0.3. If a compressive strain of 4×10^{-3} is produced in a uniform rod of this material, the percentage change in its volume will be

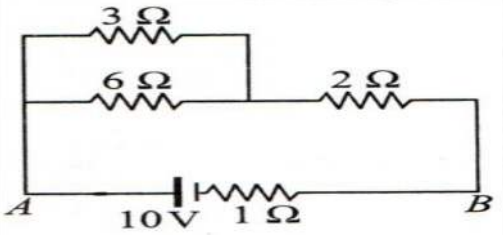
1) Increases by 0.24% **2)** Decreases by 0.24%

3) Increases by 0.16% **4)** Decreases by 0.16%

39. A solid conducting sphere, having a charge Q, is surrounded by an uncharged conducting hollow spherical shell. Let the potential difference between the surface of the solid sphere and that of the outer surface of the hollow shell be V. If the shell is now given a charge of $-4Q$, the new potential difference between the same two surfaces is:

1) $-2V$ **2)** $2V$ **3)** $4V$ **4)** V

40. The ratio of terminal p.d to emf is



1) 0.2 **2)** 0.8 **3)** 1.2 **4)** 1.5

41. A single slit of width b is illuminated by a coherent monochromatic light of wavelength λ. If the second and fourth minima in the diffraction pattern at a distance $D=1m$ from the slit are at $3cm$ and $6cm$ respectively from the central maximum, what is the width of the central maximum? (i.e. distance between first minimum on either side of the central maximum)

1) $1.5\,cm$ **2)** $3.0\,cm$ **3)** $4.5\,cm$ **4)** $6.0\,cm$

42. The electric field of an electromagnetic wave in free space is represented by $\vec{E}=E_0\cos(\omega t-kz)\hat{i}$. The corresponding magnetic induction vector will be

1) $\vec{B}=(E_0C)\cos(\omega t-kz)\,\hat{j}$ **2)** $\vec{B}=\left(\frac{E_0}{C}\right)\cos(\omega t-kz)\,\hat{j}$

3) $\vec{B}=E_0\cos(\omega t+kz)\,\hat{j}$ **4)** $\vec{B}=-\left(\frac{E_0}{C}\right)\cos(\omega t-kz)\,\hat{j}$

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43. A simple telescope, consisting of an objective of focal length $f_o = 60\,cm$ and a single eye lens of focal length $f_e = 5\,cm$ is focused on a distant object in such a way that parallel rays emerge from the eye lens. If the object subtends an angle of 2^0 at the objective, the angular width of the image is

1) 10^0 **2)** 24^0 **3)** 50^0 **4)** $(1/6)^0$

44. An electron in a hydrogen atom makes a transition $n_1 \to n_2$, where n_1 and n_2 are the principal quantum numbers of the two states. Assume Bohr model to be valid.

	Column-I		Column-II
A)	The electron emits an energy of $2.55eV$	P)	$n_1 = 2, n_2 = 1$
B)	Time period of the electron in the initial state is eight times that in the final state	q)	$n_1 = 4, n_2 = 2$
C)	Speed of electron becomes two times	r)	$n_1 = 5, n_2 = 3$
D)	Radius of orbit of electron becomes $4.77A^0$	s)	$n_1 = 6, n_2 = 3$
		t)	$n_1 = 8, n_2 = 4$

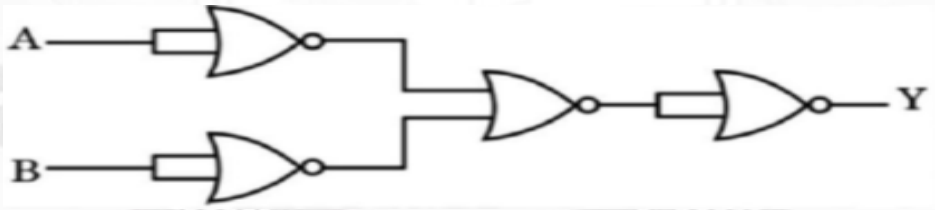
1) $A \to p; B \to p,t; C \to p,q,t; D \to r$

2) $A \to p; B \to p,t; C \to p,q,s,t; D \to r$

3) $A \to q; B \to p,q,s,t; C \to p,q,s,t; D \to r,s$

4) $A \to r; B \to p; C \to p,q,s,t; D \to q$

45. The truth table for digital circuit shown figure is:



1)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

2)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

3)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

4)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

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SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. A pulley of radius $1.5\ m$ is rotated about its axis by a force $F = (12t - 3t^2)N$ applied tangentially (while t is measured in seconds). If moment of inertia of the pulley about its axis of rotation is $4.5\ kg\ m^2$, the number of rotations made by the pulley before its direction of motion is reversed, will be $\frac{3K}{\pi}$. The value of K is .

47. Initially a gas of diatomic molecules is contained in a cylinder of volume V_1 at a pressure P_1 and temperature 250K. Assuming that 25% of the molecules get dissociated causing a change in number of moles. The pressure of the resulting gas at temperature 2000 K, when contained in a volume $2V_1$, is given by P_2. The ratio $\frac{P_2}{P_1}$ is

48. The disintegration energy Q for the nuclear fission of $^{235}U \rightarrow {}^{140}Ce + {}^{94}Zr + n$ is _____ MeV. (Given atomic masses of $^{235}U = 235.0439u; {}^{140}Ce = 139.9054u,\ {}^{94}Zr = 93.9063u,\ n = 1.0086u,$ and value of $c^2 = 931$ MeV/u.)

49. The maximum kinetic energy of a photoelectron is E when wavelength of incident light is $\frac{\lambda}{2}$. If maximum kinetic energy becomes $2E$ when wavelength is reduced to $\frac{\lambda}{3}$. If the work function of the metal is w then the value of 2w is (in eV)$\ldots\ldots\left(\frac{hc}{\lambda} = 7.5\ eV\right)$

50. A coil of self-inductance 0.01 H and resistance 1Ω is connected to $220V, 50\ Hz$ AC supply. If time lag between maximum alternating voltage and maximum current is $'k'$ second, then the value of 1000 k is…. (in sec) $(\tan 72^o = 3.14)$

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CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. If the magnetic moment of a dioxygen species is 1.73 BM, it may be

1) O_2, O_2^{-2} **2)** O_2^{+1}, O_2^{-1} **3)** O_2, O_2^{-1} **4)** $O_2^{-1}, O_2^{+1}, O_2^{+2}$

52. In a fuel cell methanol is used as fuel and oxygen gas is used as an oxidizer. The reaction is $CH_3OH_{(l)} + \frac{3}{2}O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)}$

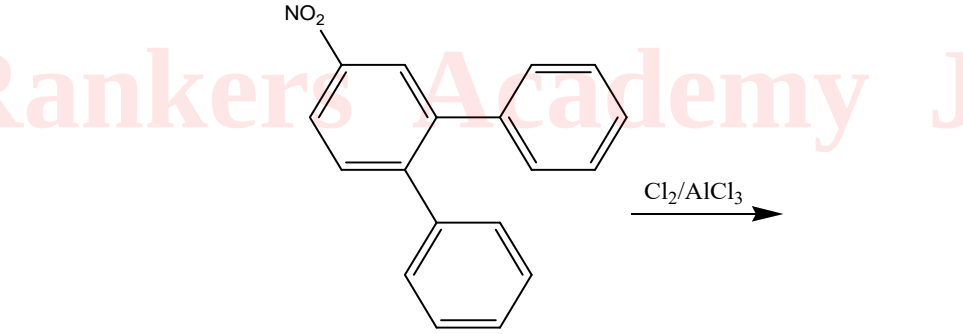
At 298 K standard Gibb's energies of formation for $CH_3OH_{(l)}, H_2O_{(l)}$ and $CO_{2(g)}$ Are $-166.2. -237.2$ *and* $-394.4\ kJ\ mol^{-1}$ respectively. If standard enthalpy of combustion of methanol is $-726\ kJ\ mol^{-1}$, efficiency of the fuel cell will be

1) 80% **2)** 87% **3)** 90% **4)** 97%

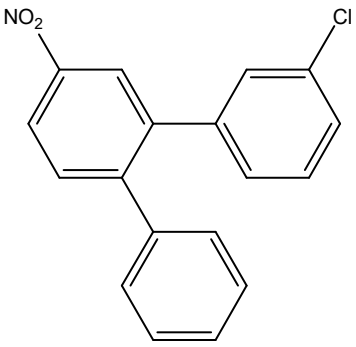
53. The solubility of P_2Q_3 is 'a' mol / lit. Its solubility product is

1) $6a^4$ **2)** $64a^4$ **3)** $36a^5$ **4)** $108a^5$

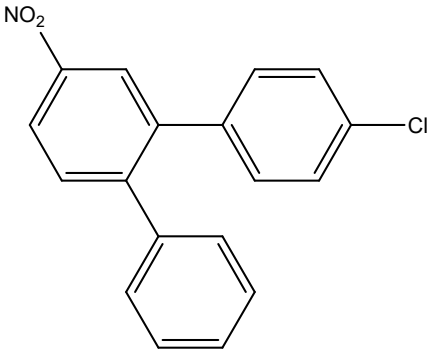
54. The major substitution product of the following reaction is



1)



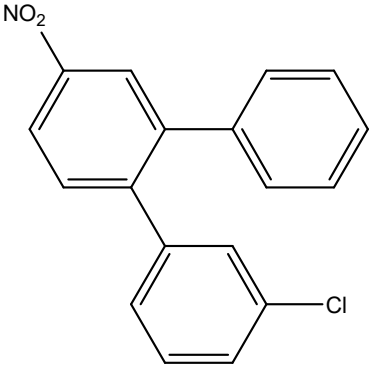
2)



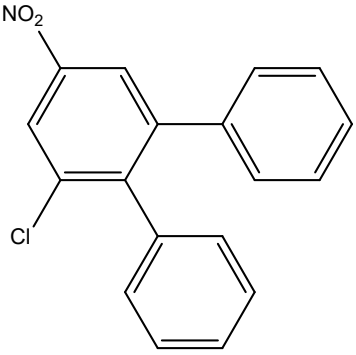
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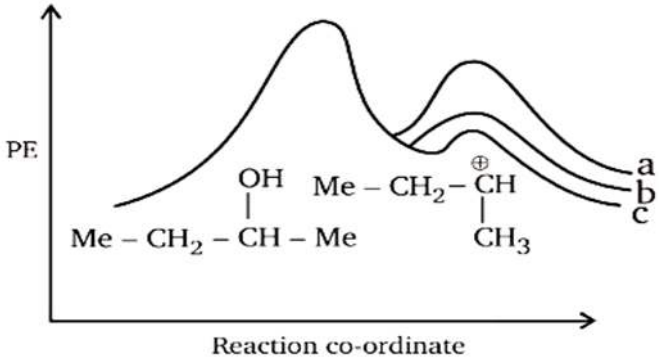


3)



4)

55. Energy profile diagram for dehydration of 2-butanol using conc. H_2SO_4 is given below:



Product (b) of above reaction is:

1) 1-butene **2)** cis-2-butene **3)** trans-2-butene **4)** iso-butene

56. Statement – 1: $CH_3-CH_2-Cl+NaI \xrightarrow[\Delta]{Acetone} CH_3-CH_2-I+NaCl\downarrow$

Statement – 2: Acetone is polar-protic solvent and solubility order of sodium halides decreases dramatically in order $NaI > NaBr > NaCl$. The last being virtually insoluble in this solvent and a 1° and 2° chloro alkane in acetone is completely driven to the side of Iodoalkane by the precipitation reaction.

1) Statement-1 is true, Statement-2 is true and Statement-2 is correct explanation for statement-1.

2) Statement-1 is true, Statement-2 is true and Statement-2 is Not the correct explanation for statement-1.

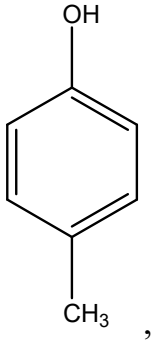
3) Statement-1 is true, Statement-2 is false.

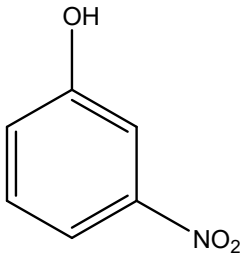
4) Statement-1 is false, Statement-2 is true.

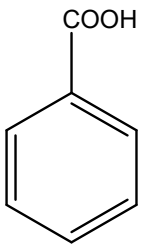
57. How many following compounds more acidic than phenol.

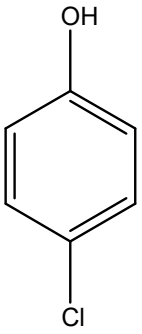
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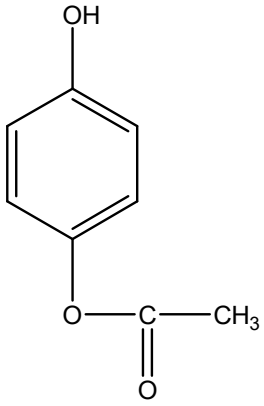

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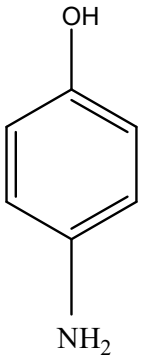


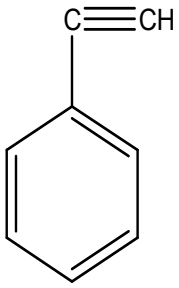
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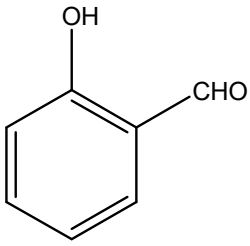
,


,
HCOOH ,




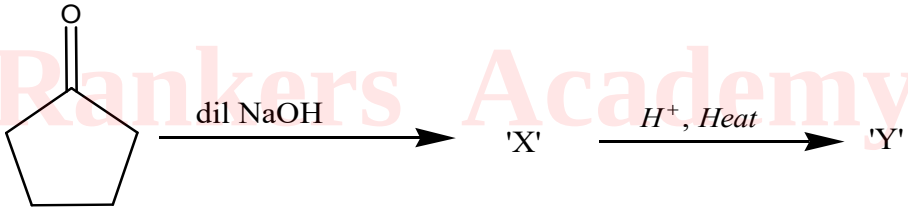
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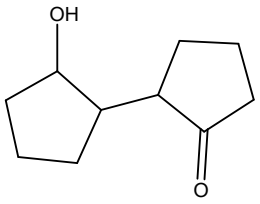
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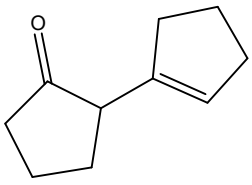
1) 4 **2)** 5 **3)** 3 **4)** 6

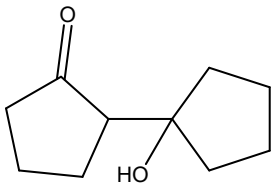
58.

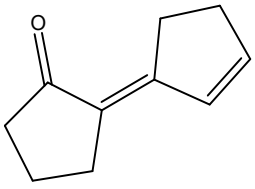


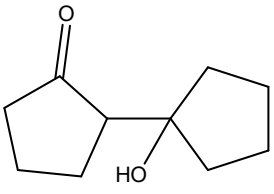
Consider the above reaction, the products 'X' and 'Y' respectively are

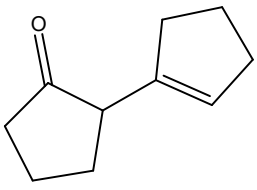
1)


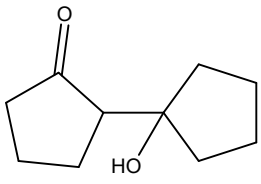
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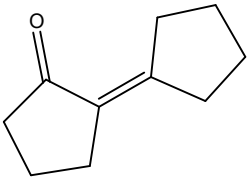
2)


,


3)


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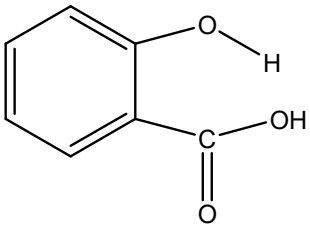
4)


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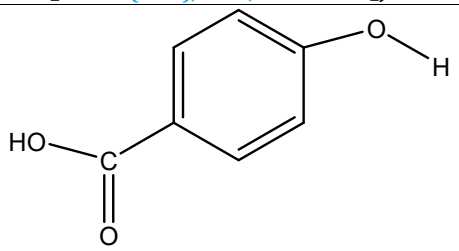
59. Consider the following molecules and statements related to them:

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A)



B)

i) (B) is more likely to be crystalline than (A)

ii) (B) has higher boiling point than (A)

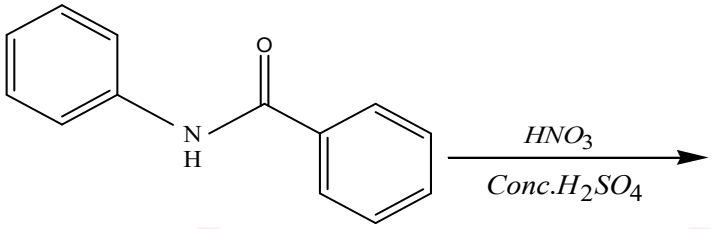
iii) (B) dissolves more readily than (A) in water

Identify the correct option from below:

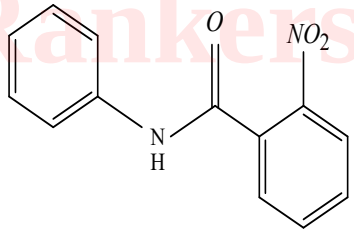
1) (i) and (ii) are true. **2)** (i) and (iii) are true.

3) All are true. **4)** (ii) and (iii) are true.

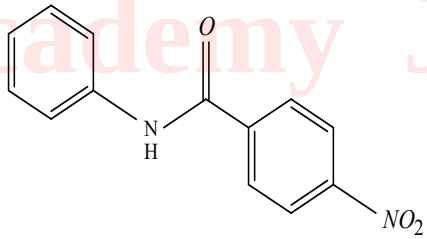
60. What will be the major product in the following mononitration reaction?



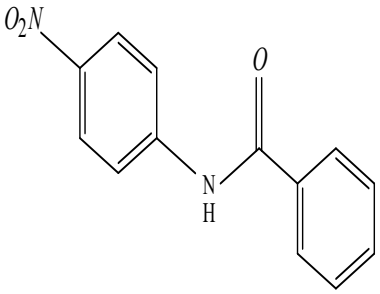
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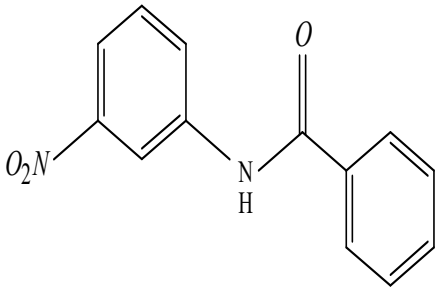
1)



2)



3)



4)

61. Match List – I with List – II.

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List – I (Reactions)		List – II (Reagents)	
A	$CH_3(CH_2)_5-\underset{\underset{O}{\|}}{C}-OC_2H_5 \rightarrow CH_3(CH_2)_5CHO$	I	CH_3MgBr, H_2O
B	$C_6H_5COC_6H_5 \rightarrow C_6H_5CH_2C_6H_5$	II	$Zn(Hg)$ and con.HCl
C	$C_6H_5CHO \rightarrow C_6H_5CH(OH)CH_3$	III	$NaBH_4, H^+$
D	$CH_3COCH_2COOC_2H_5 \rightarrow CH_3\underset{\underset{H}{\|}}{C}(OH)CH_2COOC_2H_5$	IV	$DIBAL-H, H_2O$ -78^{0}C

Choose the correct answer from options given below:

1) A-(III), (B)-(IV), (C)-(I), (D)-(II) **2)** A-(IV), (B)-(II), (C)-(I), (D)-(III)

3) A-(IV), (B)-(II), (C)-(III), (D)-(I) **4)** A-(III), (B)-(IV), (C)-(II), (D)-(I)

62. An oxide of a non – metal has the following properties:

i) Acts both as a proton donor as well as proton acceptor

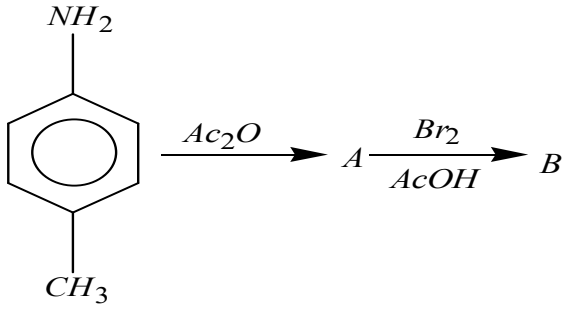
ii) It reacts readily with basic and acidic oxides

iii) It forms oxide with Fe at its boiling point.

The oxide is:

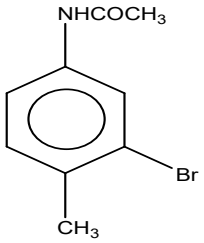
1) P_2O_5 **2)** SiO_2 **3)** H_2O **4)** CO_2

63. In the following reaction sequence,

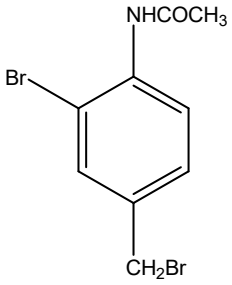


The major product B is

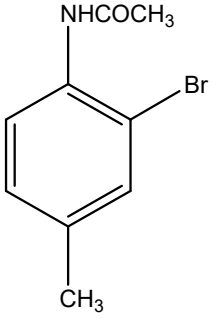
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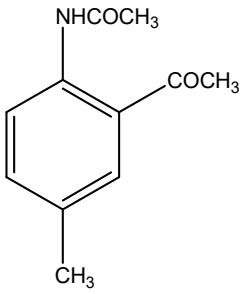
1)



2)



3)



4)

64. Statement – I: The A,D,E,K vitamins are fat soluble vitamins.

Statement – II: Deficiency of vitamin – B_1 causes for Beriberi decease.

1) Statement-I is true; Statement-II is false. **2)** Statement-I is false; Statement-II is true.

3) Statement-I & Statement-II both are true. **4)** Statement-I & Statement-II are false.

65. $\left[Fe(H_2O)_6\right]^{+2}$ and $\left[Fe(CN)_6\right]^{-4}$ differ in

1) Geometry, magnetic moment.

2) Geometry, hybridization.

3) Magnetic moment, Colour

4) Oxidation state, number of electrons in t_2g orbitals.

66. Assertion : Aquated copper (I) cation undergoes disproportionation as:

$$2Cu^{+}_{(aq.)} \rightarrow Cu^{2+}_{(aq.)} + Cu_{(s)}$$

Reason : Hydration energy of Cu^{2+} is higher than that Cu^{+} which compensates second Ionization energy of Cu.

1) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

2) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

3) Assertion (A) is true but reason (R) is false.

4) Assertion (A) is false but reason (R) is true.

67. From the given:

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	Experiment	Observation
SALT (X)	$\xrightarrow[H_2S]{NH_4OH+NH_4Cl}$	Curdy PPT
	$MnO_2 + con\,H_2SO_4$	Greenish Yellow gas
	$dil\ HNO_3 + AgNO_{3(aq)}$	White PPT
	$NH_4OH_{(aq)} + NH_4Cl_{(s)} + H_2S$	Dirty white PPT
	$K_4[Fe(CN)_6] + dil\ HCl$	Bluish White PPT

The salt (X) is

1) $ZnCl_2$ **2)** $ZnBr_2$ **3)** $MnCl_2$ **4)** $NiCl_2$

68. Which of the following is incorrect?

1) Relative lowering of vapour pressure is independent of the nature of the solute and the solvent.

2) The vapour pressure is a colligative property.

3) Vapour pressure of a solution is lower than that of the solvent.

4) The relative lowering of vapour pressure is depends upon number of particles of solute only in a dilute solution .

69. For a gaseous reaction, following data is given:

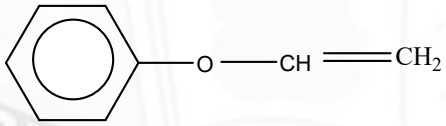
$A \rightarrow B, k_1 = 10^{15} e^{-2000/T}$

$C \rightarrow D, k_2 = 10^{14} e^{-1000/T}$

The temperature at which $k_1 = k_2$ is:

1) 1000 K **2)** 2000 K **3)** 868.82 K **4)** 434.2 K

70. Assertion : Does not undergo cleavage with HI



Reason : Neither phenylic nor vinylic cation is stable.

1) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

2) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

3) Assertion (A) is true but reason (R) is false

4) Assertion (A) is false but reason (R) is true

SECTION-II

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(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. At a certain temperature, the half change period for the catalytic decomposition of ammonia were found as follows:

Pressure (Pascals):	6667	13333	26666
Half - life period in hours:	3.52	1.76	0.88

the order of reaction is_____. $(\log 2 = 0.3010)$

72. The number of waves made by a Bohr electron in an orbit of maximum magnetic quantum number 3 is:

73. The number of acidic oxide in the following.

$NO_2, CO, SO_3, ZnO, Cl_2O_7, MnO, Sc_2O_3, PbO, B_2O_3, BeO, Al_2O_3, NO, CrO_3$.

74. The given reaction $FeS_2 + O_2 \rightarrow Fe_2O_3 + SO_2$.

The n factor for FeS_2 in the reaction is.

75. The E^o at $25^o C$ for the following reaction at the indicated concentrations is $1.5V$.

ΔG in kJ at $25^o C$ is $-X\ kJ$ .Then X *is* ____

$Cr(s) + 3Ag^+(aq, 0.1M) \rightarrow Ag(s) + Cr^{3+}(aq, 0.1\ M)$

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